

Hybrid photovoltaic and wind mini-grids in Kenya: Techno-economic assessment and barriers to diffusion

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ABSTRACT

Mini-grids have been widely developed for rural electrification purposes in Kenya, predominantly as systems combining photovoltaic modules (PV), a diesel generator, and battery capacity. (Small) wind turbines have been largely neglected in such setups and are basically non-existent in Kenyan mini-grids despite potential complementarity with PV due to different temporal production profiles of PV and wind power.

This study investigates why small wind turbines have largely been omitted from Kenyan mini-grids from both a techno-economic and diffusion theoretical perspective. A techno-economic model is used for backtesting the feasibility of a hybrid PV/wind system, and building upon theories of innovation and diffusion of technology a conceptual framework is developed and applied for assessing the state of the Kenyan mini-grid sector and categorizing identified barriers accordingly.

The article thus suggest an interdisciplinary approach to investigating the potential role of small wind power in energy systems transitioning towards renewables.

The techno-economic modelling shows that PV/wind hybrids have both technical and economic potential at average wind speeds above 4.5 m/s but little relevance below. Comprehensive barriers within the technical, economic and systemic spheres are encountered by established stakeholders of the Kenyan mini-grid sector, severely hindering diffusion of hybrid PV/wind solutions.

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1. Introduction

In Kenya, a large share of the population is without access to electricity and the benefits provided by this. Figures from 2018 state an electrification rate of 64.5% which furthermore is unevenly distributed, thus in rural areas the share is even smaller (USAID, 2018). To increase electrification a variety of technological solutions are needed, including mini-grids which in recent years have emerged as an alternative to grid expansion and solar home systems (Hansen, Pedersen, & Nygaard, 2015; Pedersen, 2016). Mini-grids could, as opposed to solar home systems, be capable of producing enough electricity to allow for development activities (Azimoh, Klintenberg, Wallin, Karlsson, & Mbohwa, 2016), and the average lumen and lighting hours are also significantly higher for grid-electrified households (Groth, 2019). However, to realise the potential of mini-grids for electrification, continued efforts to negate financial, governance, and organisational challenges are

needed (Abdallah, Bressers, & Clancy, 2015; Bhattacharyya & Palit, 2016).

Electrification and mini-grids appear to be of high interest and priority from both the private and governmental sector in Kenya (Pedersen & Nygaard, 2018, Ministry of Energy, 2018). The Rural Electrification and Renewable Energy Corporation (RREC) is established as a government authority by the Ministry of Energy, in charge of rural electrification in Kenya. The current strategic plan aims at ensuring access to electricity for all Kenyans by 2022 (Ministry of Energy, 2018).

Electrification of the rural communities is supposed to be done primarily by mini-grids developed by RREC, who as of March 2019 are in the process of developing 25 mini-grids, all expectedly developed as typical PV and diesel generator-based mini-grids supplemented by battery storage (Rural Electrification and Renewable Energy Corporation, 2018). In addition to these, an additional 121 future areas have been identified through the Kenya Off-grid Solar Access Project (K-OSAP) which are supposed to be developed and commissioned by 2020 (Rural Electrification Authority, 2018). This shows that there is a large interest in developing mini-grids, at least from a political and governmental

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List of acronyms

CRF	Capital recovery factor
Dgen	Diesel generator
HOMER	Hybrid optimization of multiple energy resources
IREK	Innovation and Renewable Electrification in Kenya
K-OSAP	Kenya off-grid solar access project
KPLC	Kenya Power and Lighting Company
LCOE	Levelised cost of electricity
NOCT	Nominal operating cell temperature
NPC	Net present cost
PV	Photovoltaic
REREC	Rural Electrification and Renewable Energy Corporation (former REA)
SWT	Small wind turbine
TAC	Total annualized cost
USD	United States dollar

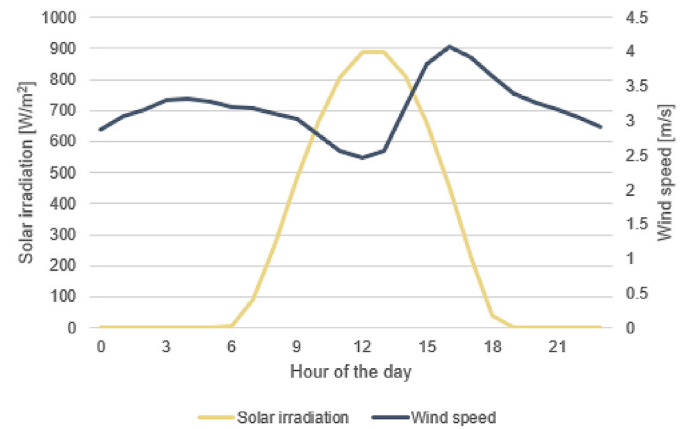


Fig. 2. Daily average resource profile for solar irradiation and wind speed for case location in Kenya. Location –0.422, 33.750, island in Homabay County. Based on hourly ERA5 data from 2017 at 10 m hub height (Copernicus Climate Change Services, 2018).

point of view.

The combination of PV and diesel generators has been a dominant system configuration for mini-grids in Sub-Saharan Africa (Azimoh, Klintonberg, Mbohwa, & Wallin, 2017) and Kenya (Hansen et al., 2015), which makes sense given the powerful solar resource in Kenya of 5–7 kWh/m²/day (Oloo, Strobl, & Olang, 2016; Solargis,

is wind power.

From Fig. 1 it can be observed that wind speeds vary quite significantly throughout Kenya, ranging from 3 m/s to upwards of 9 m/s in the windiest regions. The windier areas are well above normal requirements for wind power exploitation.

Kenya has become successful in developing large-scale wind power projects, and potential for expansion remains

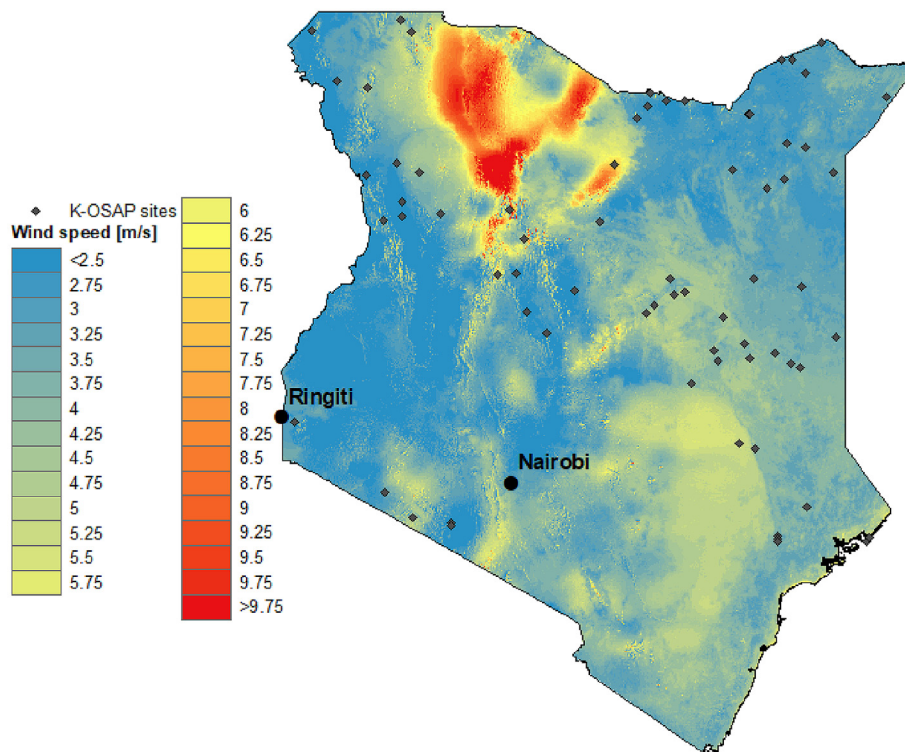


Fig. 1. Average annual wind speeds in Kenya at 12 m hub height¹ Wind speed data available at Global Wind Atlas (DTU and Global Wind Atlas, 2018).

2018). However, despite its wide utilisation, there are some obvious limitations to the use of solar power. During the night, solar power is unavailable, and electricity is typically supplied by either battery capacity or diesel generators. Diesel however is both costly, imported, and non-sustainable. A potential candidate for replacement

¹ Wind speed is recalculated from 50 m according to the wind logarithmic law, and a surface roughness length (Z_0) of 0.01 m (HOMER Energy, 2019). Wind logarithmic law: $U_{h1} = \frac{\ln(h1/Z_0)}{\ln(h2/Z_0)} U_{h2}$, where U = wind speed at hub height, $h1/h2$ = hub height 1 and 2.

(Kazimierczuk, 2019). In north-west Kenya, the Lake Turkana region, wind speeds are very high and constitute excellent areas for development of wind power, as evidenced by recent large-scale projects (Lake Turkana Wind Power, 2018). In the western and north-eastern regions wind speeds range from 2 to 4 m/s, while central and south-east Kenya have higher wind speeds of 5–6.5 m/s. Whether these low and medium wind speeds are enough to make hybrid PV/wind mini-grids relevant is largely unknown.

Looking at the average daily resource profile for solar irradiation and wind speed illustrated in Fig. 2, solar and wind complement each other well. Solar irradiation is available about 10 h every day in a very predictable pattern while wind is often available also at night.

It is these differences in availability and thus production from PV and small wind turbines (SWTs) that potentially can be exploited in Kenyan mini-grids and make hybrid PV/wind mini-grids relevant.

Electricity produced outside of the hours with available solar resource is of greater utility to the local society since it would otherwise have to be supplied by expensive batteries or diesel generators. As it is apparent from Fig. 3, there is also a demand for electricity outside of the typical production hours for PV. Supplementing PVs with one or more SWTs could potentially result in a reduced need for diesel generators, and a reduced capacity need for PV and batteries.

An additional benefit of SWTs is that SWTs potentially could be manufactured locally, as evident from the Kijito windpump experiences (Harries, 2002, 2009). This is opposed to PV modules, which in developing countries, including Kenya, are increasingly manufactured in China (Shen & Power, 2017). Thus, increased utilisation of SWTs could provide both economic and deployment related benefits. SWTs remain a niche-technology in the context of Kenya, and innovation and institutional changes are required to increase SWT deployment for rural electrification (Kamp & Vanheule, 2015; Wandera, 2018).

1.1. Existing research on hybrid mini-grids

Precisely what constitutes a mini-grid is not universally agreed upon in the literature. This paper applies the definition outlined in Franz et al. (Franz, Peterschmidt, Rohrer, & Kondev, 2014), defining mini-grids as electricity grids in remote locations at a capacity below 10 MW. Micro-grids are similar but operate at lower capacities of 1–10 kW. Common for both mini-grids and micro-grids is that they operate away from the national grid and provide the people, institutions and production facilities in close proximity with electricity. Other solutions for electrification include stand-

alone systems mostly based on diesel generators, solar home systems (up to 150 W_p), and pico-PV systems (up to 10 W_p) (Franz et al., 2014). The smaller stand-alone solutions are beneficial for electrification of very scattered households but can only provide limited electrical power thus limiting the amount and type of appliances to be connected.

The designation hybrid mini-grid is applied in this context to describe hybrid power systems in which renewable energy technologies are combined, in this case, PV and wind power. Battery storage and a backup diesel generator can be included, but this is not a requirement.

Extensive research on barriers for development and diffusion of PV and mini-grids has been conducted in the context of developing economies and Sub-Saharan Africa with typical challenges being related to; financing, viable and affordable business models, existence of public-private partnerships, and the institutional arrangements (Chaurey, Krithika, Palit, Rakesh, & Sovacool, 2012; Glemarec, 2012; Knuckles, 2016; Ugulu, 2019; Ulsrud, Rohrer, Winther, Muchunku, & Palit, 2018). Common to most existing research on mini-grids for electrification is that it has been focused on combinations of PV, diesel generators and battery capacity.

Hybrid mini-grid solutions have not gained much traction as technological configurations despite being around for years already (Fortis Wind Energy and Independent Energy, 2015; Fortis Wind Energy and Juwi Renewable, 2008). The successes from large-scale wind power projects might not be transferable to small-scale wind power due to significant differences in innovation and diffusion dynamics for small and large scale wind (and solar) projects (Hansen, Gregersen, Lema, Samoita, & Wandera, 2018), indicating a need for research on the specific application of wind power in small-scale (mini-grid) systems.

Literature on hybrid PV/wind mini-grids is available in the form of feasibility studies or techno-economic optimization, with less focus on the key stakeholders, the surrounding institutional framework and barriers to diffusion. Techno-economic modelling was used in a study investigating the potential for using hybrid PV/wind mini-grids to power a remote cell tower in Ethiopia (Yeshalem & Khan, 2017), but the findings are not applied outside of the single case study. Examples of techno-economic modelling or optimization of hybrid PV/wind mini-grids in areas in developing countries include but are not limited to assessments in Nigeria (Olatomiwa, Mekhilef, Huda, & Sanusi, 2015), Malaysia (Hossain, Mekhilef, & Olatomiwa, 2017), and India (Lal, Bhusan Dash, & Akella, 2011).

Hazelton et al. (Hazelton, Bruce, & MacGill, 2013) reviews benefits and risks of PV/diesel mini-grids for rural electrification, but since this is not conducted for a specific country, it does not include context-specific issues such as political or regulatory challenges.

The challenges, opportunities and future role of SWTs have only to a limited extent been investigated in existing research (Leary et al., 2018). SWTs and their application in rural electrification have been investigated in the context of Peru, Nicaragua, and Scotland by Sumanik-Leary (Sumanik-Leary, 2013). In this, key factors leading to success or failure of SWTs are determined, and a framework is developed to analyse the surrounding socio-technical system. The Wind Empowerment community has conducted market assessments for SWTs in Ethiopia (Eales et al., 2016), Malawi (Sumanik-Leary et al., 2016), and Nicaragua (Sumanik-Leary et al., 2012). While literature on SWTs evidently exists, the need for additional research and market assessments is a common theme in the previously mentioned examples.

1.2. Scope and structure of paper

Hybrid PV/wind mini-grids have so far been largely disregarded as a viable solution in Kenya. Whether that is a result of technical, economic or institutional barriers is unknown.

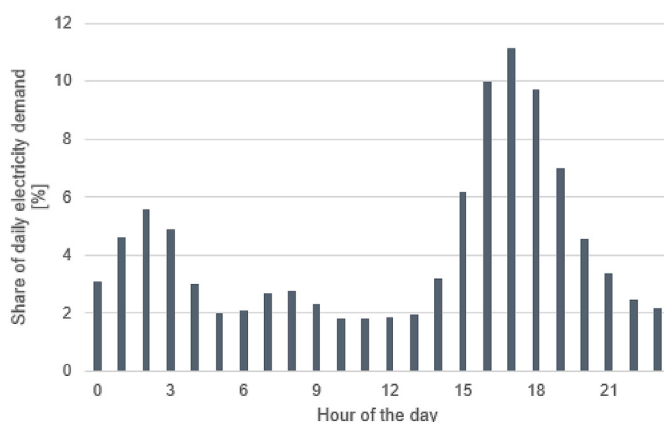


Fig. 3. Load profile for a typical African village (Prinsloo, Dobson, & Brent, 2016). Utilised for energy system modelling.

This paper sheds light on the diffusion of hybrid PV/wind mini-grids by answering the following research questions:

1. How does the complementarity of PV and wind power influence the relevance of hybrid PV/wind mini-grids in Kenya?
2. What are the barriers to diffusion of hybrid PV/wind mini-grids in Kenya?

The research questions are answered through a combination of quantitative and qualitative methodologies, an approach recommended by Schäfer et al. (Schäfer, Kebir, & Neumann, 2011) to tackle the multifaceted research topic of decentralized energy supply in developing countries. The purpose of combining the two methodological approaches is to cross-validate the data and findings and ultimately provide a more thorough and coherent picture than a single-method approach.

The quantitative and case dependent analysis in Section 4, consists of technical and economic assessment of a hybrid PV/wind mini-grid solution. Three criteria are considered important to the process of case delimitation.

1. Remote location and thus no expected future connection to the national electricity grid.
2. Mix of residential housing and (small) businesses.
3. The presence of a developer with a preliminary interest in pursuing hybrid PV/wind solutions if they prove applicable.

Based on these criteria, the case of Ringiti is investigated, an island located in the far western part of Kenya near the border to Uganda. Approximately 3200 people live permanently in 320 households, and in April 2018 a PV/battery mini-grid capable of supplying roughly half of the households with electricity was installed on the island (Renewvia Energy, 2018a). In addition to regular households, consumers include small video halls, bars and restaurants, churches, a school, barbershops, and cold drink shops.

This study backtests the feasibility of a hybrid PV/wind solution on Ringiti. The aim of the assessment is not as such to provide a detailed analysis of this single case but rather to utilise techno-economic modelling to investigate the complementarity of wind and solar resources and its viability in Kenyan mini-grids.

The qualitative and case-independent part of the paper, Section 5, involves an investigation of barriers for diffusion of hybrid PV/wind mini-grids as perceived by key stakeholders involved in the Kenyan mini-grid sector. Building upon two main theoretical concepts, Innovation and Diffusion theory, hybrid mini-grids are considered as more than physical artefacts and simple technological inventions but are also considered in light of the surrounding organisational and institutional system. This calls for a need to study a variety of processes involved in technological change including national policies, networks of organisations and institutions; a highly holistic view and approach, recommended for investigating technological change in developing countries (Heeks & Stanforth, 2015). To do so, a conceptual framework was developed and is presented in Section 2.2.

The key findings from both the quantitative and qualitative analysis are discussed with the purpose of determining what constitutes the most important barriers for diffusion of hybrid PV/wind mini-grids and to what extent hybrid PV/wind mini-grids are relevant in the context of Kenya.

The novelty of this study revolves around the applied interdisciplinary approach, combining quantitative methods in the form of detailed hourly simulations of mini-grid configurations, with the use of qualitative methods to gather and analyse empirical data. This provides policy-makers, mini-grid developers, and researchers with an up-to-date status of the Kenyan mini-grid sector, and an assessment of the potential for hybridizing mini-grids with SWTs.

2. Theories and methods

This chapter first presents existing literature on diffusion and innovation theory in the context of renewable energy and developing economies. From this, a conceptual framework relevant to this specific study is constructed and presented in Section 2.2. The main qualitative and quantitative research methods and their application and relevance to this study are then presented in Section 2.3 and 2.4

2.1. Diffusion and innovation theory in renewable energy

Diffusion of technology and innovation is an extensive research field with a nearly endless amount of different theoretical conceptualisations, with perhaps the most widely utilised being Rogers (Rogers, 2003). To help explain different rates of adoption of innovations, Rogers presents five characteristics of innovation as perceived by individuals: **1.** Relative advantage, **2.** Compatibility, **3.** Complexity, **4.** Trialability and **5.** Observability (Rogers, 2003). These characteristics have been widely used to research and describe processes of innovation diffusion but are typically applied to describe how individuals adopt new inventions in western economies.

Rogers' theory on diffusion of innovations has been applied in a variety of research fields, including environmental- and renewable energy technologies (Aizstrauta, Ginters, & Eroles, 2015; Dibra, 2015; Franceschinis et al., 2017) but despite its generally broad application in research it has not been utilised much in a context of developing economies and is thus not tailored to the challenges met there (Zanello, Fu, Mohnen, & Ventresca, 2015).

Rogers presents an argument for how technology perception concerning perceived advantages, compatibility and complexity influences diffusion of technology among individuals. This limits its applicability in the context of diffusion of mini-grids. Mini-grids can be considered a collectively owned asset, and in most cases, adoption is not an individual decision but requires a community to reach an agreement. Furthermore, as described in Section 1.1 rural electrification is impeded by barriers such as financing and affordability, technology management and know-how, and institutional arrangements, which are not prominent in the five characteristics of innovation. This results in a need to diversify the understanding of diffusion of innovation and technology in the context of developing economies and rural electrification to understand the potential and barriers for diffusion of hybrid mini-grids fully.

Prahalad et al. (Prahalad, Di Benedetto, & Nakata, 2012) presents a more context-specific framework for investigating innovation in bottom of pyramid markets. Bottom of pyramid markets are characterised by consisting of upwards of four billion people living on less than US\$2/day (Prahalad et al., 2012). Prahalad et al. suggest that innovation can be realised through the four A's: Awareness, Access, Affordability, and Availability. This introduces new key parameters which are not present in Rogers' framework but have the potential to greatly influence the diffusion of innovation and technology in developing economies where capital may be scarce, or technologies, knowledge and capacity is not as widely available as in developed economies. Similarly to Rogers, Prahalad et al. does not include a systemic level and instead focus on diffusion among individuals.

Barnett (Barnett, 1990) presents a further specified context and framework for investigating the diffusion of energy technologies in rural areas in developing countries, identified as nine key issues: **1.** The role of market and state, **2.** Diffusion strategies and participation, **3.** The performance of technology, **4.** Users' needs, **5.** Political economy, **6.** Financial returns, **7.** The macro policy environment, **8.** The system overhead costs, and **9.** Adaptation in the light of

experience (Barnett, 1990). In these Barnett highlights how the role of non-market forces, interactions and pressure from governmental, non-governmental and foreign aid influence the diffusion of technology. The involvement of both private and public sector actors can further complicate or facilitate the diffusion process.

Similarly to Prahalad et al., a stronger emphasis on financial returns for the user and affordability is present in Barnett's framework compared to Rogers'. Furthermore, the importance of the political environment and influence of macro-level politics on micro-level technical change and community needs is emphasised in Barnett's framework and can be considered a representation of a systemic level not present in Rogers' or Prahalad et al.

From these three different theoretical frameworks, it is clear that diffusion of innovation and technology can be approached in different ways. Ultimately, a researcher's choice of conceptual framework will depend equally on what technology is investigated and what societal context the technological innovation is to be diffused into. Furthermore, based on Section 1, it is also clear that developing economies face specific challenges that may not be present in general diffusion frameworks. Keeping in mind the context of Kenya will be important when investigating barriers for diffusion technological innovations. This calls for an augmentation of these existing frameworks to better suit the specific context and research question of this study.

2.2. Conceptual framework

The conceptual framework presented in Fig. 4 is partly based on the existing theoretical frameworks for diffusion of innovation and technology by Rogers (Rogers, 2003), Prahalad et al. (Prahalad et al., 2012), and Barnett (Barnett, 1990), but is adjusted based on the researcher's experiences from engaging with the Kenyan mini-grid sector and a preliminary categorization of the collected dataset.

The barriers for diffusion of hybrid mini-grids are considered present in three major spheres: Systemic, Technical, and Economic barriers. All these are partly present in the frameworks presented by Rogers, Prahalad et al., and Barnett, which form the foundation

of the conceptual framework, or in other words, "view of the world", applied in this study.

The barriers included in Fig. 4 originate from very different areas to provide a holistic understanding of the potential for diffusion. A similar approach for categorization of barriers is applied in Painuly (Painuly, 2001), in which a framework for assessment of barriers to increased penetration of renewable energy technologies is developed, validating such a qualitative approach for barrier assessment. The following subsections will elaborate on how the three major spheres systemic, technical and economic barriers are defined.

2.2.1. Systemic barriers

The systemic barriers can be categorized as three key issues: Political environment, State of the market, Community needs and acceptance.

The political environment includes current governmental goals, strategies, and policies. This influences prioritization and management of electrification projects. Policies can include both micro-level policies which are typically technology-specific, and macro-level policies that on a more general level influence the diffusion of technology (Barnett, 1990; Muchunku, Ulsrud, Palit, & Jonker-Klunne, 2018; Negro, Alkemade, & Hekkert, 2012).

The current state of the market revolves around how the market for mini-grid development functions and what actors are involved. The extent to which the market is a free market, a monopoly or something in-between, including the ease of which private actors can enter the market, are key issues and greatly influenced by the existing actors and the political environment (Jacobsson & Johnson, 2000; Kebede & Mitsufuji, 2017; Negro et al., 2012; Sergi et al., 2018).

Community needs and acceptance represents the challenges of adapting the innovation, in this case, mini-grids, to the local conditions and needs. This is an essential part of mini-grid development, and as previously described, the process of diffusion is not so much about individual technology adoption as it is about community adoption, which is why this is included as a systemic barrier. In some cases, it might not even be a community decision, but

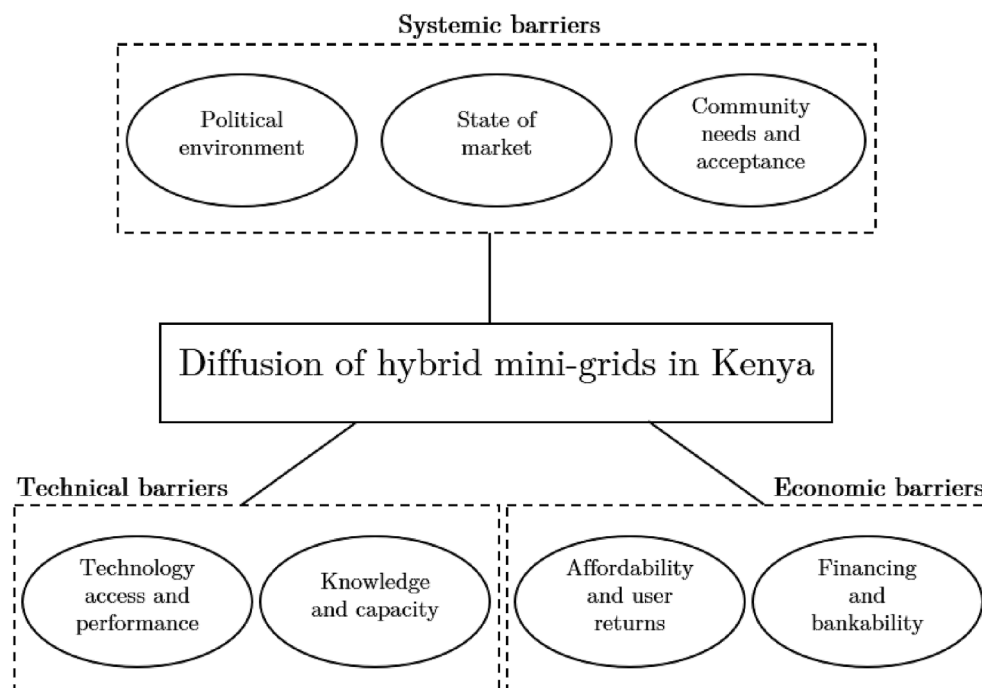


Fig. 4. Conceptual framework serving to guide the identification of barriers for diffusion of hybrid PV/wind mini-grids in Kenya.

instead a result of the current political environment and state of the market (Jacobsson & Johnson, 2000).

2.2.2. Technical barriers

Technical barriers are categorized as two key issues: Technology access and performance and Knowledge and capacity.

Technology access and performance relates to the application and function of the technology, how it compares to existing alternatives and how it performs on key technical and economic parameters (Barnett, 1990; Prahalad et al., 2012).

Knowledge and capacity consist of the local know-how and technical skills needed for implementation and operation. This is not included in the three diffusion theory frameworks discussed in Section 2.1, but for this specific use case and context, it is considered important and thus included in the contextual framework (Kebede & Mitsufuji, 2017; Negro et al., 2012).

2.2.3. Economic barriers

Economic barriers are categorized as two key issues: Affordability and user returns and Financing and bankability.

Affordability and user returns is about matching ability- and willingness to pay by making the cost of electricity sufficiently affordable to the users (Barnett, 1990; Muchunku et al., 2018; Prahalad et al., 2012).

Financing and bankability on the other hand is about ensuring a profitable business case and allowing investors to recoup the investment in a timely manner (Muchunku et al., 2018).

2.3. Qualitative barrier analysis

Qualitative research has the potential to provide key insights, explanations of complex behaviour, and system understanding, all crucial in the process of public policy development. The purpose is to address questions and identify parameters such as: what is the nature of people's experiences, critical elements of operation within the system, factors underlying attitudes and perception, barriers to system operation, actions or services needed to improve programmes and system, strategies to overcome newly defined problems (Huberman & Miles, 2002).

The qualitative data set consists of three primary data sources: 1) literature from existing projects and case studies on both local, national and international level, 2) discussions and interviews with stakeholders, and 3) site visits to relevant locations. This methodology aligns well with the framework for assessment of barriers to increased penetration of renewable energy technologies developed by Painuly (Painuly, 2001).

To comprehensively determine barriers a variety of generated data is needed, thus interviews were conducted to supplement desk research and document analysis. The primary data consist of formal and informal interviews with key industry stakeholders related to the Kenyan mini-grid sector including private firms and developers, public organisations, research institutions and NGOs. A total of seven formal interviews and five informal interviews have been carried out in the period of August 2018 to November 2018.

Interviews were conducted with the purpose of gathering empirical data from key stakeholders, needed to identify and categorize barriers to diffusion of hybrid PV/wind mini-grids. Thus, interview questions revolved around uncovering what experiences the interviewees have with hybrid PV/wind solutions, in addition to how the potential of such solutions is perceived.

Formal interviews have been conducted as semi-structured interviews based on interview guides, recorded, transcribed and imported to NVivo (NVivo, 2019). Informal interviews are, as implied, informal discussions that were not recorded, and the resulting data consists of notes made during and after the discussions and then imported to NVivo. NVivo was used both as a tool to

assist in maintaining an overview of the collected data and to organize and categorize the data according to theme and case using a system of nodes.

The process of categorization started with only two nodes: 1) Barriers for mini-grids, and 2) Barriers for hybrid mini-grids. As the data was analysed this developed into a list of 21 hierarchical nodes, which form the foundation of the barriers presented in Section 5.

The semi-structured interview approach was chosen due to its flexibility and thus compliance with the exploratory nature of the study. A very strict interview guide or survey-based approach relies more heavily on the existing knowledge of the researcher for developing this preceding framework and there is a risk that important topics are overseen. The semi-structured approach allows the researcher to pursue topics that appear most critical and important to the interviewee (Kvale, 1996).

Table 1 lists the interviewees included in the seven formal interviews conducted. For discretionary reasons and to ensure the anonymity of interviewees, names and organisational roles are excluded. The organisations mentioned in Fig. 1 were represented by one or two employees highly involved in the Kenyan mini-grid sector.

In addition to the conducted interviews, the data set consists of field notes from site visits to two existing mini-grids (Ringiti island and Ndeda island) and two nano-grids (Echariria and Lomolo B) in Kenya. Furthermore, participation and resulting discussions from the Future Energy East Africa Conference, and participation in a stakeholder meeting organised by the Kenya Mini-wind Project (Nygaard et al., 2018), in which 34 key stakeholders from the Kenyan mini-grid sector were present.

Qualitative data is, by nature, unstructured and unwieldy due to it consisting of a variety of text-based sources including transcriptions of interviews, discussions, and field notes. The contents are detailed and in micro-form, e.g. observations, descriptions, and personal experiences. Bringing structure to this range of data is a challenge and an important part of qualitative research (Huberman & Miles, 2002). In this study, the conceptual framework developed in Section 2.2 functions as a map to guide and revise the research design and ultimately broaden the situational understanding.

2.4. Energy system analysis using the HOMER model

The purpose of the energy system analysis is to determine under what conditions hybrid PV/wind mini-grids are preferable to typical diesel-only or PV/diesel generator/battery mini-grids.

To simulate and optimise energy system configurations, the software HOMER (Hybrid Optimization Model for Multiple Energy Resources) is utilised, a tool capable of combining engineering and economic aspects (HOMER Energy, 2018). HOMER was originally developed for use in off-grid hybrid systems and has later been expanded to be compatible with grid-connected systems. HOMER allows for simulation and optimization of all possible system configurations including sensitivity parameters, making it suitable for assessment of hybrid PV and wind mini-grids. HOMER is a widely acknowledged tool applied in several previous instances for assessment of mini-grid systems in sub-Saharan Africa (Azimoh et al., 2017; Olatomiwa, Mekhilef, Huda, & Sanusi, 2015; Yeshalem & Khan, 2017).

Several tools exist for simulating and optimizing energy systems. Studies have previously reviewed and compared an extensive list of existing tools including EnergyPLAN, EnergyPRO, TRNSYS16 and RETScreen with regards to simulation possibilities and previous use cases (Connolly, Lund, Mathiesen, & Leahy, 2010; Lyden, Pepper, & Tuohy, 2018). The purposes of these tools range from analysing national and regional energy systems, local and community-scale systems, or single plants and projects. Time step options range from 0.01 s to daily with project horizons upwards of

Table 1

Formal interviews conducted in Kenya in the period of September 2018–November 2018.

Organisation	Description
KCIC	Innovation centre, provides support and capacity building to Kenyan entrepreneurs within climate and clean energy technologies.
REREC	Primary governmental entity responsible for rural electrification strategies, incl. K-OSAP project.
I-DEV International	Business strategy and advisory firm.
Strathmore ERC	Managing partner of green mini-grid facility. Research institution specialized in renewable energy in Kenya and East Africa.
Solargen	Quality assurance of PV modules.
Renewvia Energy	Private mini-grid developer.
Powergen	Private mini-grid developer.

50 years. HOMER is chosen for this study due to its focus on modelling local and community-scale projects and its ability to determine optimal system configurations through technical and economic optimization for many technologies and varying energy resources. The default simulation time step in HOMER is 1 h, but can range from 1 min to several hours.

2.4.1. Economic parameters

A discount rate of 14% and an inflation rate of 4.5%², resulting in a real discount rate of 9.09% is assumed. Different economic parameters are used to evaluate and assess hybrid mini-grid configurations, these are described in the following.

Total net present cost (NPC). NPC is the present value of all costs incurred during the project lifetime minus the revenue earned. In this case, expenses include capital expenses, replacement costs, O&M costs, and fuel costs. Revenue is limited to include salvage value.

NPC is calculated as the sum of all discounted cash flows.

$$NPC = \sum_{t=0}^n NP_t \cdot (1+i)^{-t} > 0 \quad (1)$$

where:

NPC: Total Net Present Cost

n: Lifetime

i: Discount rate

NP_t: Net payment or expenses to time t

NPC is used to rank the system configurations and used for calculation of total annualized cost (TAC) and levelised cost of electricity (LCOE).

Salvage value. Some production units have expected lifetimes shorter than the project horizon of 25 years and thus replacements will be needed. Replacements are included at the same capital cost as the initial investments. The value of components with remaining lifetime at the end of the 25-year project horizon will be subtracted from the NPC at salvage value assuming a linear depreciation.

$$S = C_{rep} \cdot \frac{R_{rem}}{R_{comp}} \quad (2)$$

where:

S: Salvage value

C_{rep}: Replacement cost

R_{rem}: Remaining lifetime

R_{comp}: Component lifetime

Levelised cost of electricity (LCOE). LCOE is defined as the average cost per kWh of useable electricity. This is calculated based on the total annualized cost of the system divided by the electric load served.

$$LCOE = \frac{TAC}{E_{served}} \quad (3)$$

where:

LCOE: Levelised cost of electricity

TAC: Total annualized cost

E_{served}: Total delivered electricity

TAC is the annualized value of the NPC, calculated as the capital recovery factor (CRF) multiplied the NPC.

$$TAC = CRF(i, n) \cdot NPC \quad (4)$$

The CRF is calculated as a function of the discount rate and the project lifetime

$$CRF(i, n) = \frac{i(1+i)^n}{(1+i)^n - 1} \quad (5)$$

3. Model description and assumptions

A schematic overview of a hybrid mini-grid system can be seen in Fig. 5. Typically, a generator and battery storage are included to negate the intermittency and fluctuations of renewable energy resources and secure a constant supply of electricity. An inverter is included to allow a flow of electricity between DC and AC components. When renewable energy production exceeds the electricity demand, the battery will be charged and can later be discharged when renewable electricity production is low.

In this study, a project horizon of 25 years is assumed, and the energy system analysis is conducted in hourly time-steps where for every hour the electricity demand is supplied by either the available production units or the battery. Hourly time-steps have become tradition in studies focusing on the planning and feasibility stage of projects, for example studies conducted with the tool EnergyPLAN (Lund, 2014; Marczinkowski & Østergaard, 2019; Thellufsen, Nielsen, & Lund, 2019; Østergaard, 2015). Furthermore, larger systems, such as mini-grids, are capable of balancing demand and production to a greater extent than very small systems such as individual households where shorter time-steps could be needed even for the planning phase.

² The average 12-month inflation rate for January to October of 2018 in Kenya (Central Bank of Kenya, 2018).

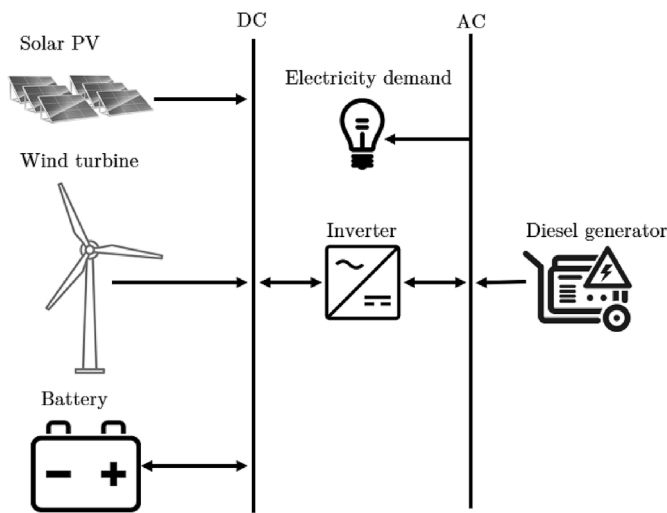


Fig. 5. Schematic overview of hybrid mini-grid as analysed in HOMER.

3.1. Electricity demand

The load profile utilised is a result of a scoping study in which the hourly load profile for a typical rural African village was determined and can be observed in Fig. 3 (Prinsloo et al., 2016). This profile resembles the typical consumption pattern observed in recent Kenyan feasibility studies and demand assessment projects (African Solar Designs, 2018; Economic Consulting Associates, 2014; Innovation Energie Développement, 2018).

No seasonality of the electricity demand is included in the analysis since the technologies present on Ringiti do not have season-specific use cases. Furthermore, the equatorial location of Kenya results in low seasonal weather fluctuations and there is therefore not a significant increase in electricity demand because of heating technologies or similar.

The total electricity demand is a result of the connected users and general assumptions for electricity consumption per user based on the appliances and commercial facilities present on Ringiti.

KPLC has estimated the average consumption for users in their mini-grids to be 121 kWh/month/user. KPLC primarily targets high development areas and many of these areas have been electrified for a long time already, resulting in development and thus higher consumption.

In three different feasibility studies of mini-grids, the average electricity consumption was estimated to range from 80 to 106 kWh/month/user (Economic Consulting Associates, 2014). The sites analysed are in the range of 1000–2000 users and therefore larger than the context of this study and thus these numbers might be too high since a higher number of businesses and institutions can be expected to be present. A study on villages of similar size to the one investigated in this project study estimates the average demand to be 35–62 kWh/month/user (Carbon Africa Limited et al., 2015).

All this results in an assumption of 50 kWh/month/user. Users consist of approximately 23 commercial consumers and 87 residential consumers, resulting in an annual demand of 66 MWh and a peak demand of 20.38 kW.

3.2. Economic/technical parameters and weather data

Table 2 provides an overview of the technologies included in the model, associated capital and operational expenses, and lifetimes.

In Table 3 technical parameters for the included technologies can be seen.

The annual average wind speed on Ringiti is 3.18 m/s and the daily average solar irradiation is 6.28 kWh/m²/d. The seasonal variation on a monthly basis can be seen in Table 4. The data included for solar irradiation and wind speed is ERA5 data, a climate reanalysis dataset (Copernicus Climate Change Services, 2018). The wind speed data used for the techno-economic assessment is for a hub height of 10 m (see Table 5).

ERA5 data is available in hourly resolution and has a spatial resolution of 31 km. For solar irradiation, there are no immediate concerns with using such a dataset, but for the wind power assessment, locally measured data would be preferable because of the uncertainty involved in a data set with a high spatial resolution. Since the focus of this study is on the general complementarity effect of PV and wind it is deemed sufficient for this purpose.

In recent years MERRA and MERRA-2 data produced by NASA have been popular for modelling of wind power generation, but according to Olauson (Olauson, 2018) and EMD International (EMD International, 2018), ERA5 improves accuracy compared to MERRA and MERRA-2.

3.3. Control strategy

Two primary control strategies are included in the model, load following and cycle charging.

The load following strategy will make the generator only produce enough power to meet the current residual demand after electricity from renewable technologies and storage has been subtracted. Renewable technologies will produce to the storage when there is excess production. This is typically optimal for systems with a high share of renewable electricity.

The cycle charging strategy will make the generator operate at full capacity whenever a generator is required, and excess production will charge the storage.

Thus, the two main differences are whether partial load operation is allowed on the generator - and whether the generator may charge the battery. HOMER will test both control strategies for all system configurations and select the one that results in the lowest system cost.

3.4. Sensitivity variables

To investigate the influence of varying climate conditions and to discuss the relevance of hybrid PV/wind mini-grids in Kenya a sensitivity analysis is conducted. Only the magnitude of the wind speeds and solar irradiation changes; the temporal profile remains the same throughout the analysis.

4. Results of techno-economic assessment

HOMER simulated a total of 698524 different solutions for the study of this case with a computational time of 3 h. A solution is a combination of production and storage units, optimized according to the solar and wind resources, capital and investment costs, and the demand requirements. This means one solution could be a PV/diesel generator/battery/inverter system, and this system configuration is then ranked according to the NPC of other simulated systems.

4.1. Ringiti base scenario

Table 6 provides an overview of the six cheapest system configurations for the Ringiti base scenario.

System A1: PV/Dgen/Battery provides the lowest NPC and LCOE for the Ringiti base scenario, followed by System A2: PV/Wind/Dgen/Battery. The wind power investment in System A2 is minimal and the production is also only 1733 kWh annually equalling 1.7% of

Table 2

Cost overview for included technologies. Total investment cost is a result of the cost shown and the capacity included in each scenario. The investment cost shown includes an expense of 20% for balance of system components.

Technology	Size	Investment [USD]	O&M	Lifetime	Source
Diesel generator	23 kW	13800	0.35 USD/h	15000 h	African Solar Designs (2018)
Wind turbine	5.4 kW	15272	3%	20 years	Fortis Wind Energy (2018)
PV	1 kW	1200	1%	20 years	Moner-Girona et al. (2018)
Battery	1 kWh	455	1%	2285 kWh	African Solar Designs (2018)
Inverter	1 kW	600	—	15 years	African Solar Designs (2018)
Installation costs (PV)	1 kW	150			African Solar Designs (2018)

Table 3

Technical parameters for included technologies. ^aEfficiency of the diesel generator depends on the load it is being operated at, increasing from 25% to 35% at maximum capacity. ^bPower output per hour is calculated as a function of wind speed and the turbine specific power curve (Fortis Wind Energy, 2013).

Technology	Model	Technical parameters	
Diesel generator	Generic	Efficiency ^a	25%–35%
Wind turbine	Fortis Montana	Rated power ^b	5.4 kW
		Rotor diameter	5 m
		Hub height	12 m
PV	Jinko JKM275-60	Rated power	275 W
		Temperature coefficient	-0.11%/°C
		NOCT	45°C
		Derating factor	88%
Battery	Trojan SAGM 12 205	Nominal voltage	12 V
		Nominal capacity	2.63 kWh
		Roundtrip efficiency	85%
		Minimum charge	20%
		Lifetime	2285 kWh
Inverter	Generic	Efficiency	95%

Table 4

Solar and wind resources on Ringiti, location: -0.422, 33.750.2017 ERA5 data (Copernicus Climate Change Services, 2018).

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Solar irradiation [kWh/m ² /d]	6.92	6.87	6.74	5.87	6.09	5.90	5.72	6.19	6.22	6.37	5.98	6.55
Wind speed [m/s]	3.68	3.41	3.72	3.30	3.09	2.99	3.07	3.25	3.22	3.04	2.45	2.96

Table 5

Sensitivity parameters included in model. First data column is the base scenario for Ringiti.

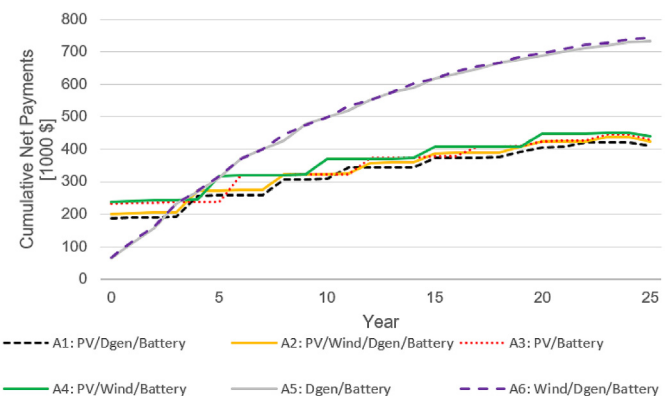
Wind speed [m/s]	3.18	3	3.5	4	4.5	5	5.5	6	6.5	7	7.5	8
Solar irradiation [kWh/m ² /d]	6.28	5	5.25	5.5	5.75	6	6.25	6.5	6.75	7		

the total energy production.

The addition of wind power in System A2 only reduces the PV capacity by 0.8 kW and has no influence on the installed PV capacity, and no influence on the total battery storage capacity compared to A1. It seems preferable to maintain a diesel generator for backup capacity equal to the peak demand, but from the very high RES when combined with a PV system it is apparent that the diesel generator does not have many production hours. Systems without diesel generators require a battery capacity that is 27–50% higher compared to systems with a backup diesel generator.

Fig. 6 illustrates the discounted cumulative costs for the six system configurations from Table 6. It can be seen how systems relying on diesel generators (systems A5/A6) for most of the production have a lower initial investment cost, but quickly become much more expensive due to increased operational expenses. The other four system configurations have similar NPC, which is due to the heavy reliance on PV for most of the production and thus investment cost.

As can be expected, the battery charges during the day and discharges during the evening. The batteries are charged during the afternoon, indicated by the red colours. During the evening and night, the batteries are discharged, indicated by the blue colours.

**Fig. 6.** Cumulative discounted costs for Ringiti base scenario.

The daily fluctuations in charge level can be seen for a year in Fig. 7.

The installed wind turbine in System A2 has 6861 operating hours, but due to the low wind speeds, the power produced is low, as illustrated in Fig. 8. For many hours during the year, the production from the turbine ranges from 0 to 0.4 kW, resulting in

Table 6

Overview of identified possible system configurations for Ringiti base scenario, 6.28 kWh/m²/d solar irradiation and 3.18 m/s average wind speed. All systems include battery storage of varying capacity and are ranked according to LCOE.

System	LCOE [USD/kWh]	NPC [1,000 USD]	Capital cost [1,000 USD]	RES [%]	Genset [kW]	PV [kW]	Wind [kW]	Battery [kWh]
A1: PV/Dgen	0.631	411	187	98.9	23	54	0	189.4
A2: PV/Wind/Dgen	0.651	425	201	99	23	53.2	5.4	189.4
A3: PV	0.658	429	233	100	0	65.7	0	284
A4: PV/Wind	0.676	440	239	100	0	72.5	5.4	242
A5: Dgen	1.12	733	73	0	23	0	0	94.7
A6: Wind/Dgen	1.14	742	79	0.1	23	0	5.4	73.6

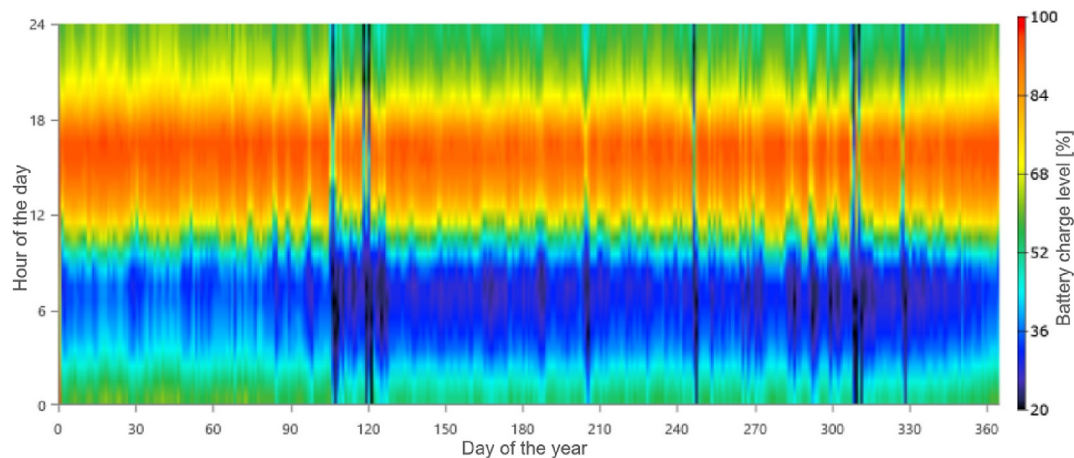


Fig. 7. Battery charge level Ringiti base scenario System A2: PV/Wind/Dgen/Battery system.

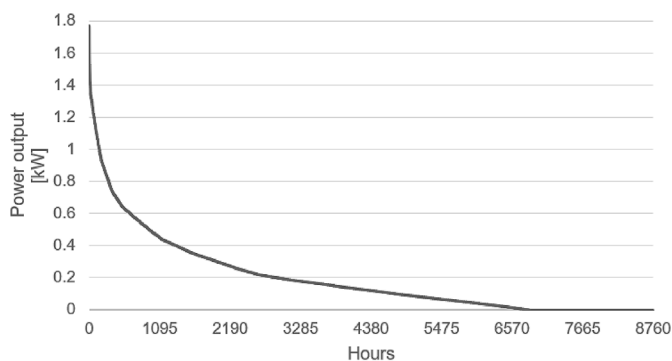


Fig. 8. Duration curve Fortis Montana 5.4 kW turbine in Ringiti base scenario System A2: PV/Wind/Dgen.

annual production equal to only 186 full load hours.

To summarize, there is little benefit to installing a SWT in the Ringiti base scenario with wind speeds averaging 3.18 m/s annually. While the low cut-in wind speed of 2.5 m/s results in a high number of operational hours, the low power content at such speeds result in low power production. The LCOE increases by 0.2 USD/kWh when including a single SWT, fails to reduce the optimal battery capacity, and only reduces installed PV capacity by 0.8 kW in systems with a backup diesel generator. For systems without a diesel generator, a SWT increases LCOE by 0.18 USD/kWh and the optimal battery capacity is reduced by 15%.

4.2. Sensitivity analysis

To determine the influence of varying solar and wind resources a sensitivity analysis is conducted. This is also to better understand the relationship between PV and wind power production in a mini-

grid setting and to determine what conditions are needed to make hybrid PV/wind mini-grids relevant to other areas in Kenya.

4.2.1. Influence of solar and wind resources

The range of variance in solar and wind resources is chosen to resemble the range in Kenya. Although not a sensible scenario for Ringiti, the analysis shows that hybrid PV/wind solutions are not sub-optimal everywhere, dependent on local solar irradiation and wind resources. Two significant thresholds for the wind speed can be observed in Figure 9 and 4.5 m/s and 6 m/s. For wind speeds below 4.5 m/s, the optimal system configuration is a PV/Dgen/Battery system, but at higher wind speeds, it is beneficial to include wind power as well. For wind speeds higher than 6 m/s a diesel generator can be excluded, and the optimal system solution relies solely on PV and wind power production.

It is worth noting that in no situation is it beneficial to have a system without PV even at lower solar irradiation levels, and how battery capacity is included in all optimal system configurations.

The results from Fig. 9 illustrate how PV/wind hybrids seemingly are the optimal solution at sufficiently high wind speeds. To further illustrate this the results from a high wind speed scenario of 5.5 m/s are included in Table 7. SWTs have in other off-grid systems been implemented in areas with similar wind resources (Bhandari et al., 2014; Nayar, Tang, & Suponthana, 2008).

In a situation with a higher average wind speed of 5.5 m/s, including wind power is preferable. LCOE of System B1: PV/Wind/Dgen is 0.046 USD/kWh lower than System B3: PV/Dgen without wind power. Comparing systems B1 and B3 it can be seen that battery capacity can be reduced by 11% and PV capacity by 29% in System B1, but this reduction does not entirely offset the increased capital cost from investing in SWTs.

For System B2: PV/Wind compared to System B4: PV, both without diesel generators, LCOE can be reduced by 0.064 USD/kWh, battery storage by 29%, and PV capacity by 19% when including

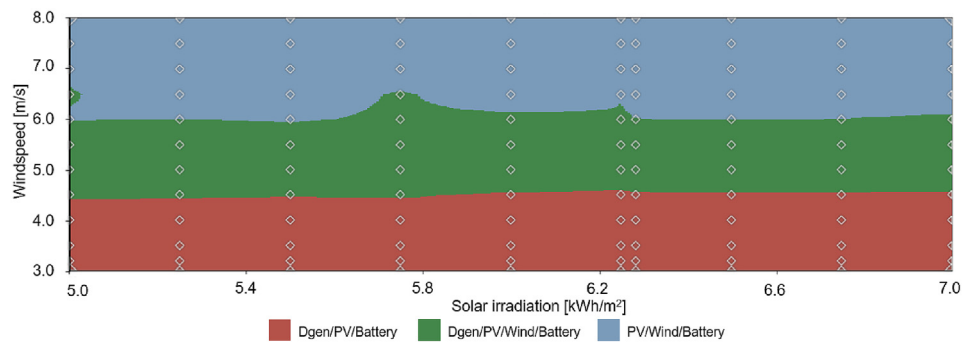


Fig. 9. Optimal general system topologies at solar irradiation levels ranging from 5 to 7 kWh/m²/d and wind speeds ranging from 3 to 8 m/s. Specific capacities for included technologies are not shown. The colours indicate what system topology is optimal under the given wind speed and solar irradiation levels. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 7

High wind speed scenario, 6.28 kWh/m²/d solar irradiation and 5.5 m/s average wind speed. All systems include battery storage of varying capacity.

System	LCOE [USD/kWh]	NPC [1000 USD]	Capital cost [1000 USD]	RES [%]	Genset [kW]	PV [kW]	Wind [kW]	Battery [kWh]
B1: PV/Wind/Dgen	0.585	382	216	99.3	23	37.9	21.6	168.3
B2: PV/Wind	0.594	387	238	100	0	53.2	21.6	199.9
B3: PV/Dgen	0.631	411	187	98.9	23	54	0	189.4
B4: PV	0.658	429	233	100	0	65.7	0	284
B5: Wind/Dgen	0.801	523	258	78.1	23	0	64.8	105.2
B6: Dgen	1.12	733	73	0	23	0	0	94.7

wind power. From this, it seems the complementarity effect of PV and wind is much more apparent for systems without diesel generators. The large reduction in battery capacity combined with the reduced PV capacity results in System B2 having a capital cost that is only USD 4616 (2%) more than System B4.

Fig. 10 illustrates the discounted cumulative costs for different system configurations. It is clear that systems without PV are much more expensive during a 25-year project lifetime.

The annual wind power production of System B1 in a high wind speed scenario of 5.5 m/s increases to 30008 kWh. This is equal to 29.6% of the total electricity production, or 1389 full load hours.

4.3. Relevance of hybrid mini-grids in Kenya

The previously determined wind speed thresholds of 4.5 m/s and 6 m/s respectively entail changes in optimal system configuration for a mini-grid. These thresholds will be used to quantify the extent of where hybrid PV/wind mini-grids are relevant in Kenya and how this relates to the current K-OSAP project and the mini-grids expected to be developed as part of this. Fig. 11 illustrates

this correlation of wind speed thresholds, Kenyan geography, and 72 of the 121 planned K-OSAP mini-grids.

Fig. 11 divides Kenya into three different categories according to wind speed. Areas with wind speeds below 4.5 m/s are most likely not relevant for hybrid PV/wind mini-grids and are illustrated as red areas. Areas with wind speeds ranging from 4.5 to 6 m/s are

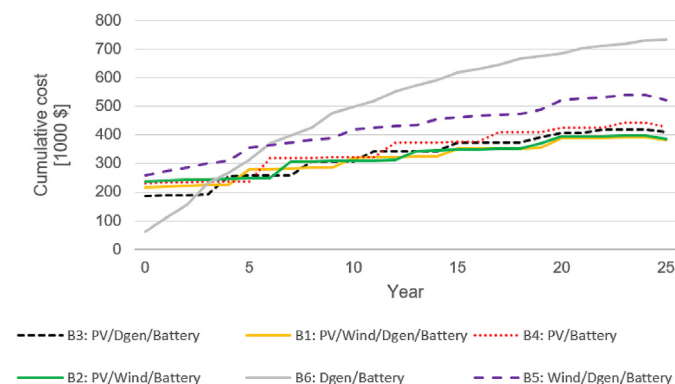


Fig. 10. Discounted cumulative costs for high wind speed scenario at 5.5 m/s average wind speed.

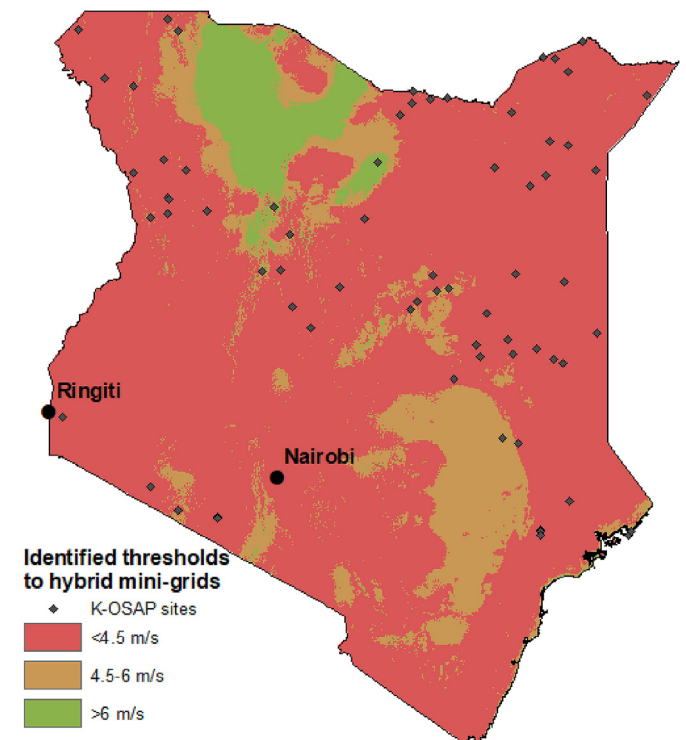


Fig. 11. Geographical overview of areas most relevant to hybrid PV/wind mini-grids. Based on the determined thresholds from Section 4.2 and wind speed data presented in Fig. 1 (DTU and Global Wind Atlas, 2018), adjusted in GIS software.

above the identified threshold for making hybrid PV/wind mini-grids relevant, indicated as orange areas. Planned sites located within these areas should be subject to further site-specific analysis to identify the potential. Areas with wind speeds above 6 m/s are above the threshold for when a hybrid PV/wind mini-grid can forego a diesel generator, and the electricity production will be split almost evenly between the two technologies, indicated as green areas. These areas should be high priority areas for hybrid PV/wind mini-grids.

Only 72 of the proposed 121 K-OSAP mini-grids are included in Fig. 11 due to a lack of coordinates for the remaining sites. From Fig. 11 it can be seen how 12 sites (17%) are located in the orange or green areas indicating that hybrid PV/wind is likely the lowest cost solution.

An important reservation to Fig. 11 is the site-specific and local nature of wind. This map is to be seen as a general assessment and not as a comprehensive decision-making tool. Wind speed will fluctuate locally and pockets with wind speeds both higher and lower than what is indicated in Fig. 11 will exist. The purpose is primarily to indicate the potential for hybrid PV/wind mini-grids on a national scale as opposed to assessment of individual sites, where site-specific assessments still should be conducted.

4.4. Key findings

- At wind speeds above 4.5 m/s hybrid PV/wind mini-grids are preferable, and above 6 m/s, the optimal system configuration omits a diesel generator.
- At wind speeds averaging 5.5 m/s hybrid PV/wind mini-grids can reduce the LCOE by 7–10% and battery storage requirements by 11–29%.
- The additional investment cost of wind power has minimal influence on the initial capital cost due to the reduction in PV and battery capacity.
- In the context of Ringiti, and most likely also neighbouring islands in Lake Victoria, hybrid PV/wind mini-grids are not preferable to typical PV/diesel generator or PV only mini-grids.
- The complementarity effect of solar and wind resources is much more prevalent and impactful in mini-grids without diesel generators.
- Of the 121 planned K-OSAP mini-grids, at least 12 show a preliminary potential for development as a hybrid PV/wind mini-grid and site-specific assessments should be conducted.

5. Qualitative barrier assessment

This section utilises the conceptual framework established in Section 2.2 to determine systemic, technical, and economic barriers. Then the expected influence and hindrance to diffusion of hybrid PV/wind mini-grids is rated to provide an immediate look at where the most critical barriers are present.

5.1. Barriers to hybrid PV/wind mini-grids in Kenya

It is perhaps in this context of mini-grids not surprising how most issues relate to the introduction of the new technology: wind power. Table 8 provides an overview of the identified barriers for hybrid PV/wind mini-grids in Kenya based on the collected data described in Section 2.3.

These barriers are further elaborated and related to the conceptual framework in the following sections.

5.1.1. Systemic barriers

The political environment for electrification, in general, is positive due to the strong emphasis on electrification and the ambitious 2022, 100% access to electricity goal. However, while the

overall goal has been established, the path towards it is still unclear, leaving the sector in a state of uncertainty especially prevalent amongst wind power advocates. There are currently no policies, goals or strategies related to wind power, and critical actors such as the Ministry of Energy have no activities within SWTs or goals of quantifying the potential for PV/wind hybrids. This is exemplified by how there is currently no budget within the Ministry of Energy for SWT projects.

The current state of the mini-grid market is greatly influenced by the regulatory framework and governance system actors must operate within. Realizing how a market is not static but rather a product of several factors and parameters that can both facilitate and hinder diffusion of mini-grids is important. Currently, the mini-grid market is difficult to navigate due to the difficulty of obtaining licenses, permits, and approval of tariffs.

Furthermore, there are many different interests at stake in the Kenyan electricity sector. Kenya Power and Lighting Company (KPLC) and KenGen effectively have a monopoly on electricity distribution and production and seemingly have little interest in providing a framework and market for independent power producers. As such, the market for mini-grids is perhaps not an entirely free market and realizing the potential of hybrid mini-grids can be expected to be difficult due to the uneven and unregulated playing field that is the current state of market.

The three interviewed mini-grid developers all emphasise that the existing regulatory framework is lacking, and primarily focus on solar with very little to no guidelines and regulation for SWTs. Similar viewpoints were expressed by stakeholders at the Future Energy East Africa Conference and at the Kenyan Mini-wind Project stakeholder meeting.

This opinion is prevalent among most stakeholders, and it appears that the lack of standards and regulations hinders wind power development, and the requirement of conducting environmental impact assessments for wind power projects adds a level of complexity to the feasibility study, drawing on the development process.

Problems of community knowledge and awareness are also expressed by the interviewed mini-grid developers, who experienced how locals have expressed concerns with the noise from wind turbines, influence on bird and other wildlife, and sickness or reduced pregnancy rates because of the turbines. Barriers for development that are not present for PV, a technology that is well known in Kenya due to its wide application and history.

These issues are most likely not present everywhere, and the communities of Echariria and Lomolo B where solar nano-grids have been installed both expressed positive interest towards wind power. The positive outlook from the community with regards to wind power was repeated at both Ringiti and Ndeda, where the developer has already engaged the community to discuss the future possibility of implementing wind power. These site experiences do lead to the realization that obtaining community acceptance for wind power might not be as big a challenge as some developers imply and perceive.

5.1.2. Technical barriers

A large majority of the stakeholders agree that the technological knowledge and capacity for wind power is insufficient. There are very few local suppliers of SWTs. Therefore, both hardware and skilled workers would most likely have to be imported from outside of Kenya, increasing project costs. One stakeholder did express the viewpoint that technological knowledge and capacity in Kenya generally is high, as exemplified by the successfully developed hydropower mini-grids. Transferring this knowledge to hybrid PV/wind mini-grids might not be a straightforward process, and it can be argued that the technology-specific capacity for wind power is low.

Table 8
Categorization of the most commonly mentioned barriers for hybrid mini-grids.

Barrier	Explanation
Wind resource quality	Wind resource in Kenya is perceived to be insufficient.
Technological knowledge	Knowledge of wind technology and capacity for development is low.
Governance and regulation	Lack of framework for implementation of wind power. Additional regulatory requirements for wind projects.
Cost of wind projects	The cost of wind power is perceived to be higher than for PV.
Community acceptance	Understanding and acceptance of wind technology varies. Issues related to noise, bird incidents, health impact as a result of turbines.
Complexity wind projects	Wind projects are seen as more complex. Necessary data is not available.
Suitable technology	No suitable wind turbine exists. Low presence of local distributors or manufacturers.

The perceived available wind resource in Kenya represents a critical challenge for the development of hybrid mini-grids in Kenya. Wind speeds are by most stakeholders perceived to be too low to make wind power a cost-effective investment, but this is for most stakeholders an assumption and not something that is based on actual technical or economic assessments.

“Well, I could say wind is complex; it is not as easy to design wind, as it is to design solar. You can get information and irradiation from online tools and can just design your system from anywhere in the world, when it comes to wind you don't get that. And as you know, of course wind profiles change so much, the expertise is not there, so wind is not a cheap thing to construct.” (Strathmore Energy Resource Centre, 2018)

The interviewees from Strathmore ERC touch upon several key issues for integration of wind power in mini-grids. There is the issue of data availability and complexity of wind project design process, and the issue of expertise and technological knowledge that is lacking in Kenya. Renewvia Energy, a private mini-grid developer, confirms this viewpoint.

“The skill for especially wind, I have not heard of anyone who specializes in wind. People know very little about wind.” (Renewvia Energy, 2018b)

The complexity of wind projects thus appears to be a challenge that is difficult to overcome given the available technical capacity. This challenge is likely further aggravated by the lack of accurate wind speed data in Kenya, resulting in a situation where the potential for wind is generally not further investigated. The Ministry of Energy has been collecting wind speed data for more than 10 years across Kenya but is yet to do something with this or at least make it publicly available. This seems primarily to be due to a severe lack of resources and capacity within the Kenyan Ministry of Energy (Wandera, 2018).

The price decreases for PV modules experienced in recent years has made PV a more attractive technology than wind to the Kenyan mini-grid sector. It was evident during the interviews that most stakeholders are not up-to-date on current wind power prices. The majority assumed costs to be higher for wind than PV, perhaps a result of the low presence of local suppliers and distributors. It was widely agreed upon that the primary parameter of concern when choosing technological configuration is project economy and LCOE, since security of supply is typically not a problem due to battery and diesel generator capacity in mini-grids. The different diurnal production profiles of PV and wind power and thus resulting utility to the local society is generally not considered when comparing technologies.

Stakeholders also expressed concerns about the compatibility with the Kenyan context for the SWTs currently on the market. One

issue mentioned by KCIC is the low wind speed environment in Kenya, and thus a wind turbine should preferably function well under these circumstances.

Another issue mentioned is the requirement of having low maintenance costs. This was brought up by several stakeholders, one being Powergen, who before transitioning entirely to PV mini-grids tried to pursue a business of developing SWTs in Kenya. However, maintenance costs proved to be too high and customers could not afford those costs. The challenge of keeping maintenance costs low is further complicated by the typical very remote locations of most mini-grid sites and compared to PV this is pronounced due to the typically higher maintenance costs.

Battery recycling, while important from an environmental perspective, was not brought up by any of the stakeholders interviewed and has thus not been included in the overview of barriers presented in Table 8. This could be a result of the stakeholders' emphasis on project development and electrification, as opposed to end-of-life markets and recycling.

Economic barriers. All stakeholders agree that critical barriers to continued mini-grid expansion are project costs and ensuring financial returns to increase project feasibility. Therefore, cost will be an important; if not the most important parameter for which hybrid solutions must be competitive with existing solutions. The standard and barrier for entry is set by the well-known PV mini-grids, and unless hybrid solutions can improve project economy, there is a broad agreement that it will not become relevant and that alternative benefits such as diversifying the production composition or increasing the renewable energy share cannot outweigh economy as a deciding factor. This was exemplified in the interview with I-DEV International.

“But I don't see, if there is no economic argument for it, I don't think that it (hybrid PV/wind) is going to find much aggregation.” (I-DEV International, 2018)

However, on the discussion of whether this causes the pursuit of wind power integration to be a lost cause, stakeholders generally agree that there is a potential for including wind power in mini-grids, but focus on capital expense is imperative to the success and diffusion.

Since project economy has been established as a critical barrier and evaluation criteria, it is problematic that the same stakeholders are basically unaware of how a hybrid system would perform and what the value of wind power would be. Wind power is largely excluded from feasibility studies and the potential complementarity effect is unacknowledged. The decision to leave out wind power from mini-grid projects and even feasibility studies seems to be based on assumptions of increased project costs rather than actual quantitative assessments and modelling.

The assumptions of increased project costs and complexity are

likely related to the previously mentioned technical barriers involving a lack of technological knowledge and capacity locally. Increased project costs are typically expected to arise from increased soft costs, which are costs related to the engineering and feasibility stage of a project, and because of increased operation and maintenance costs of hybrid PV/wind projects.

5.2. Overview of barriers to diffusion

Upon review of the current mini-grid sector in Kenya and the opinion of key stakeholders, it is clear that diffusion of hybrid PV/wind mini-grids will be difficult and that extensive barriers are present within all three major spheres: Systemic, Technical, Economic. Fig. 12 summarizes the previous assessment of barriers in the context of the conceptual framework developed in Section 2.2.

The simplistic high to low impact on diffusion rating system applied in Fig. 12 is a result of the amount of references the specific themes were assigned during the NVivo analysis across all stakeholder interviews, as well as a qualitative assessment on the severity of the statements. While barriers are present and influential in all three spheres, perhaps the first thing one notices is the extensive and critical influence from the economic barriers. This is partly due to the high priority of this area from all stakeholders, with REREC being the one exception. From REREC's viewpoint, the decision of whether to develop a mini-grid or not does not as such rely on project economy since it will be subsidized and financed, but to other stakeholders it is the top priority. However, isolating economic barriers as a separate entity is difficult, since it is strongly influenced by barriers within both the systemic and technical

category. E.g. the regulatory framework and market conditions influences project economy, and similarly so does the technology performance and capacity for project development.

The barriers within the systemic category are very influential as well, for both mini-grids in general and hybrid PV/wind mini-grids. Despite the presence of many serious issues, these are only categorized as having a medium impact due to the positive trends and developments also happening. These include the strong focus and political attention to electrification and renewable energy as evidenced by the K-OSAP project and REREC's electrification strategy.

Extensive technical barriers influence the diffusion of hybrid PV/wind mini-grids as well, the most significant being the severe lack of technological knowledge and capacity. This is evidenced both on a project development level where the capacity to conduct the required assessments and feasibility studies is lacking, and on an operational level where the lack of skilled technician hinders the O&M of hybrid PV/wind systems.

5.3. Key findings

- The environment for diffusion of hybrid mini-grids in Kenya is overwhelmingly negative - a result of extensive systemic, technical and economic barriers.
- The so far low diffusion rates of hybrid PV/wind solutions are a result of both industry perception and technological complexity.
- Wind power is not a significant part of the current electrification agenda, decision-making processes, or feasibility studies for neither public nor private projects.

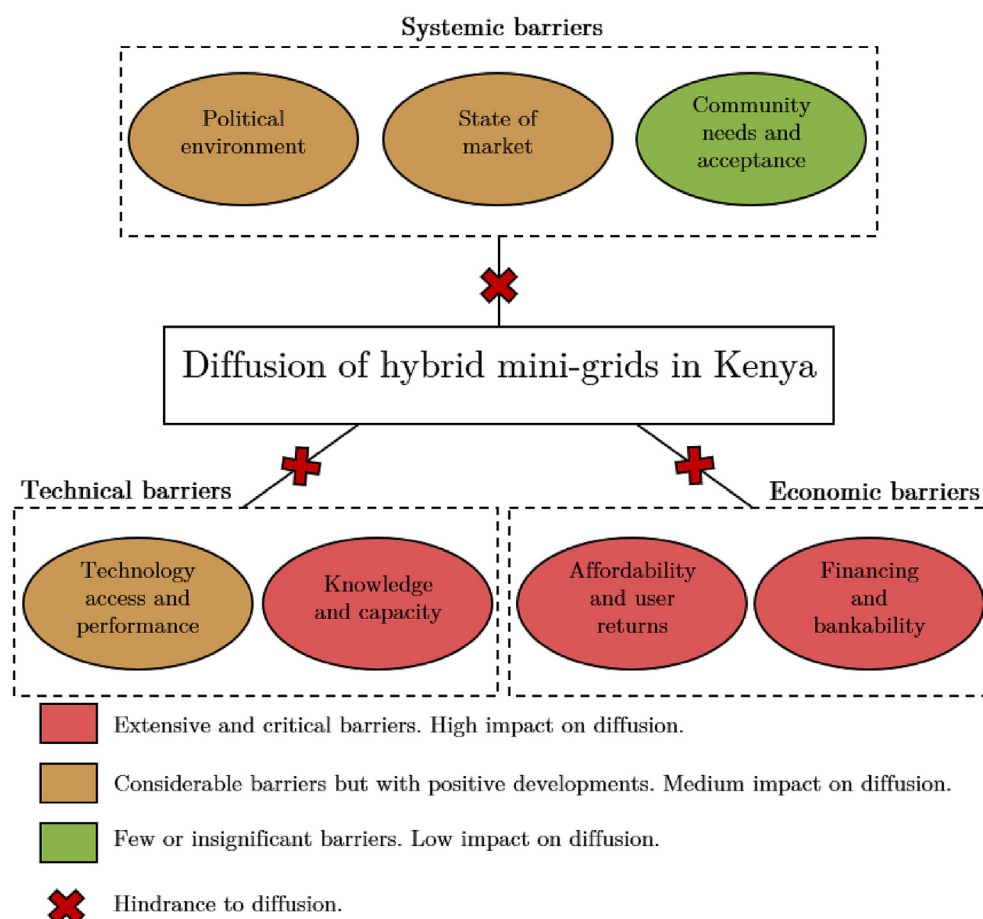


Fig. 12. Evaluation of barriers for diffusion of hybrid mini-grids in Kenya.

- The lack of regulatory framework, political goals, and strategies cause uncertainty and hinders both mini-grid in general and hybrid PV/wind mini-grid development.
- While the capacity for mini-grid development is high, the technology-specific capacity for wind power is insufficient for development of hybrid PV/wind mini-grids.
- There is a lack of up-to-date knowledge on the currently available SWT technology, costs, and accurate wind speed measurements.
- A complementarity effect of wind and PV is largely ignored and the value of hybridizing a mini-grid with wind power is unknown or unacknowledged.
- Cost is the primary parameter of interest to both public and private developers. Unless hybrid solutions can provide a similar or lower LCOE, they are largely irrelevant.

6. Conclusion

Hybrid PV/wind mini-grids show technical and economic potential to reduce LCOE in areas with average wind speeds above 4.5 m/s, but little relevance elsewhere. This makes hybrid PV/wind mini-grids relevant in several of the planned K-OSAP mini-grids, which are expected to be developed solely as PV/diesel generator/battery configurations based on very general assessments. This could be optimized, and further site-specific analyses should be conducted prior to construction.

The barriers to diffusion of hybrid PV/wind mini-grids in Kenya are extensive and the current market for mini-grid development is difficult to navigate, severely limiting the diffusion of hybrid PV/wind mini-grids. Critical barriers present in systemic, technical and economic spheres will have to be resolved before hybrid PV/wind mini-grids can be expected to have any significant contribution to electrification. The expectations are low among most key stakeholders for combining PV and wind in Kenyan mini-grids, and there is generally a lack of awareness and acknowledgement of wind power as a relevant future technology in Kenya.

In conclusion, substantial systemic, technical and economic barriers limit diffusion of hybrid PV/wind mini-grids in Kenya. While the complementarity effect of PV and wind power is largely unrecognized in Kenya, it does increase the techno-economic viability of hybrid PV/wind mini-grids, making wind power relevant as a complementary technology. Hybrid PV/wind mini-grids, while not relevant everywhere in Kenya, should be examined and developed to a greater extent than what has so far been the case.

Declaration of interest

None.

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